

Project Initialization and Planning Phase

Date	8 July 2026
Team ID	SWTID-2026-5447
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform credit card approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To Revolutionize credit card approval process using machine learning for faster, more accurate assessments.
Scope	Assesses applicant data like age, income, credit score to predict approval probability. Builds robust ML-based approval system.
Problem Statement	
Description	The current manual credit card approval process causes delays, inaccuracies, and a lack of transparency, leading to higher rejection rates.
Impact	Improves operational efficiency, reduces risk, enhances card issuance process, and boosts customer satisfaction.
Proposed Solution	
Approach	Uses a Random Forest model to predict credit card approval based on historical applicant data.

Key Features	• ML-based credit assessment model • Web-based decision-making for instant approval prediction • Continuous learning to adapt to financial trends • Transparent rejection reasons with tips to improve eligibility
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU for model training	Standard CPU
Memory	RAM for data processing	4 GB RAM or above
Storage	Disk space for data, models	500 MB - 1 GB
Software		
Frameworks	Python libraries	Pandas, Scikit-learn, Numpy
Libraries	For ML and data viz	Matplotlib, Seaborn, Flask
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Applicant data for training	Kaggle Credit Card Approval Dataset